People's Democratic Republic of Algeria
Ministry of Higher Education and Scientific Research
Tahri Mohamed University of Bechar

Fundamentals and Technique of Programming — Mini Project

شا نطيب

Sha Ntieb

An application that provides recipe recommendations based on available ingredients using a web API

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University year : 2025-2026

📋 Presentation Structure

01 🎯 **Introduction to Problem**

Theme • Problem Definition • Input/Output • Hypotheses

02 📐 **Program Design & Structure**

Flowchart • Application Logic • Main Functions

03 💻 **Source Code Structure**

Code • Data Structures • Libraries • GitHub Link

04 📸 **Results & Interpretation**

Screenshots • Execution Examples • Results

05 ⚡ **Code Enhancement**

pytest • flake8 • Performance Improvements

06 🏁 **Conclusion**

Summary • Difficulties • Areas for Improvement

Part 1 : Introduction to Problem

Theme: نظام اقتراح الوصفات الجزائرية/ **Algerian Recipe Recommendation System**

❓ Problem Definition:

Many people have ingredients at home but don't know what recipes to cook. There is no dedicated application for Algerian traditional cuisine that can intelligently suggest recipes based on available ingredients.

📥 Input Data

- List of available ingredients (Arabic/English)
- User authentication credentials
- Recipe data (via CSV or manual form)
- Rating score (1-5 stars)

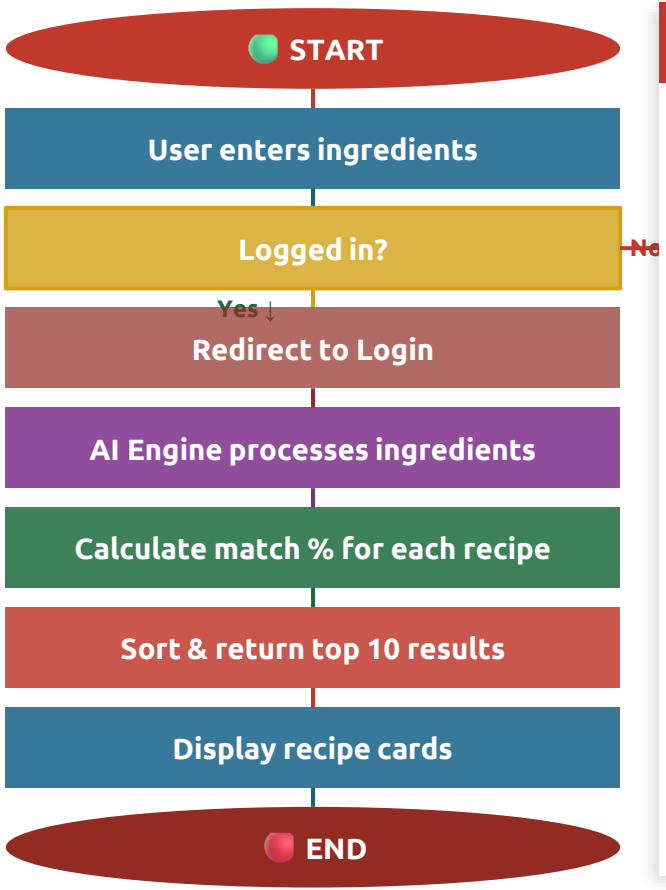
📤 Output Data

- Suggested recipes sorted by match %
- Missing ingredients for each recipe
- Recipe details (ingredients, preparation steps)
- AI similarity score per recipe

💭 Hypotheses

- User has at least 1 ingredient available
- Recipes stored in MySQL database
- AI engine works without internet
- CSV file encoded in UTF-8

Part 2 : Program Design: Flowchart



Application Logic

- **1. Input:** User types ingredients separated by commas
- **2. Auth Check:** Verify JWT token exists in localStorage
- **3. API Call:** POST /api/recipes/ai-search with ingredients list
- **4. AI Processing:** TF-IDF + Cosine Similarity calculation
- **5. Scoring:** Each recipe gets an AI score (0-100%)
- **6. Sorting:** Results sorted by score descending
- **7. Display:** Recipe cards shown with match percentage

Part 2 : Main Functions Used

⚙ normalize(text)

Normalizes Arabic/English text by lowercasing and replacing similar Arabic chars (آإأ→ا, ة→ه)

⚙ simple_match_score()

ai_engine.py

Calculates simple match between user ingredients and recipe ingredients with partial matching

⚙ tfidf_similarity()

Builds TF-IDF vectors for both ingredient lists and computes cosine similarity score

⚙ ai_recommend()

Main AI engine: combines all levels, queries DB, scores all recipes and returns sorted top-N

⚙ import_csv()

Reads uploaded CSV, detects encoding/separator, imports recipes + ingredients to MySQL

⚙ hash_password() / verify_password()

bcrypt-based password hashing and verification with 12 rounds for secure authentication

Part 3 : Source Code Structure

📁 Project Structure

```
sha-ntieb/
├── 📁 backend/
│   ├── app/
│   │   ├── api/routes/
│   │   │   ├── auth.py
│   │   │   ├── recipes.py
│   │   │   └── admin.py
│   │   ├── core/
│   │   │   ├── config.py
│   │   │   ├── database.py
│   │   │   └── security.py
│   │   ├── models/
│   │   ├── schemas/
│   │   ├── services/
│   │   │   └── ai_engine.py
│   │   └── main.py
│   └── requirements.txt
├── 📁 frontend/src/
└── recipes.csv
```

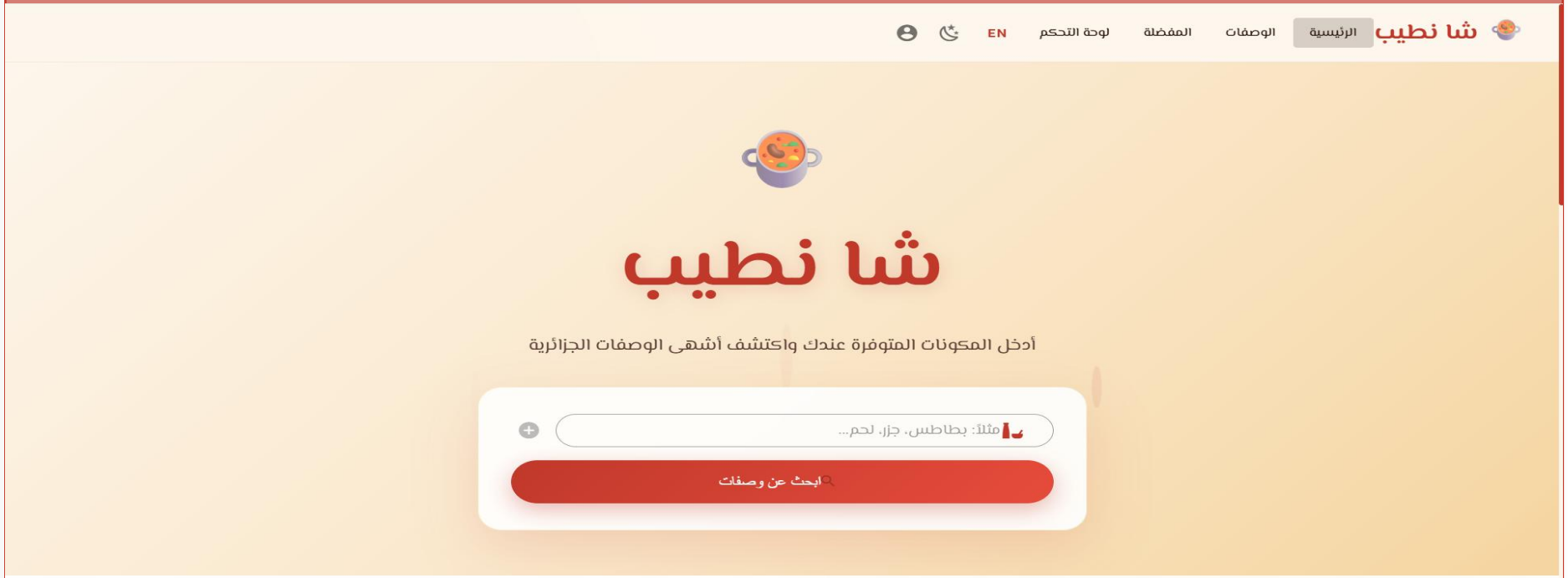
📦 Libraries Used

- 📌 **FastAPI:** REST API Framework
- 📌 **Pandas:** CSV Processing
- 📌 **python-jose:** JWT Tokens
- 📌 **SQLAlchemy:** ORM for MySQL
- 📌 **Passlib/bcrypt:** Password Hashing
- 📌 **pydantic:** Data Validation

🔗 https://github.com/feredj/sha-ntieb

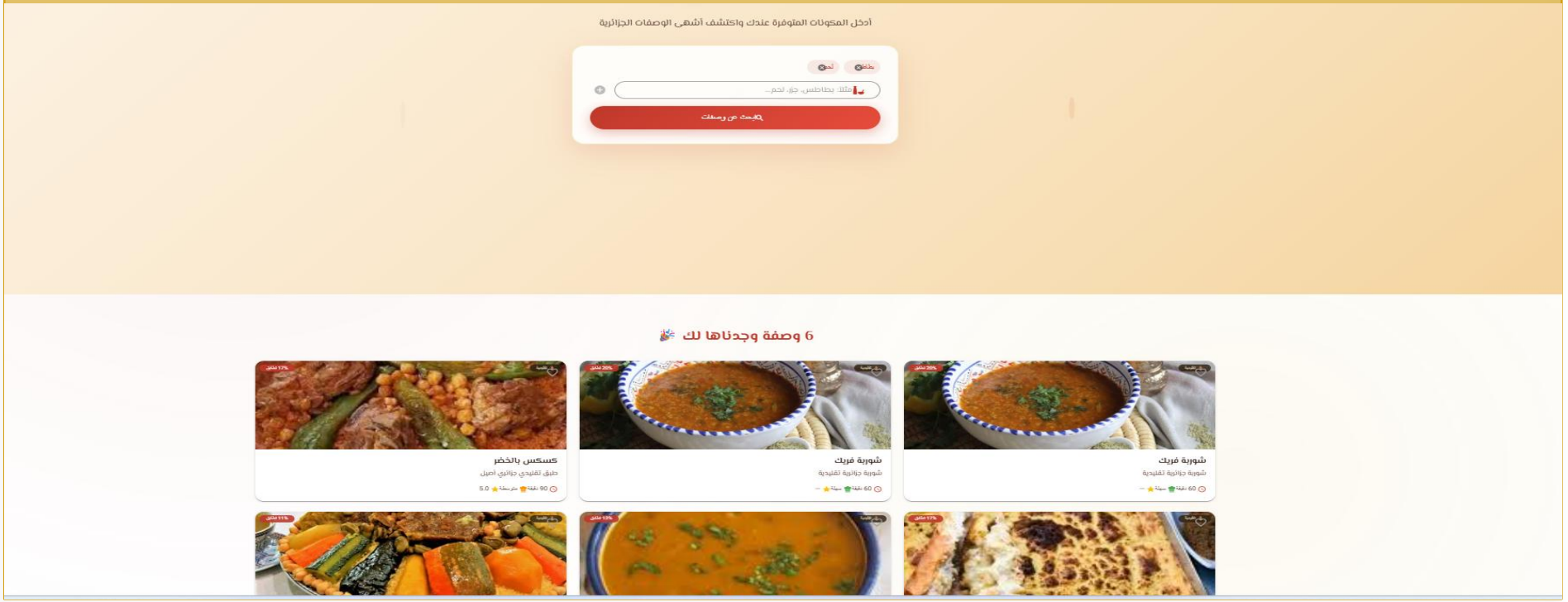
Part 4 : Results & Interpretation

Home Page

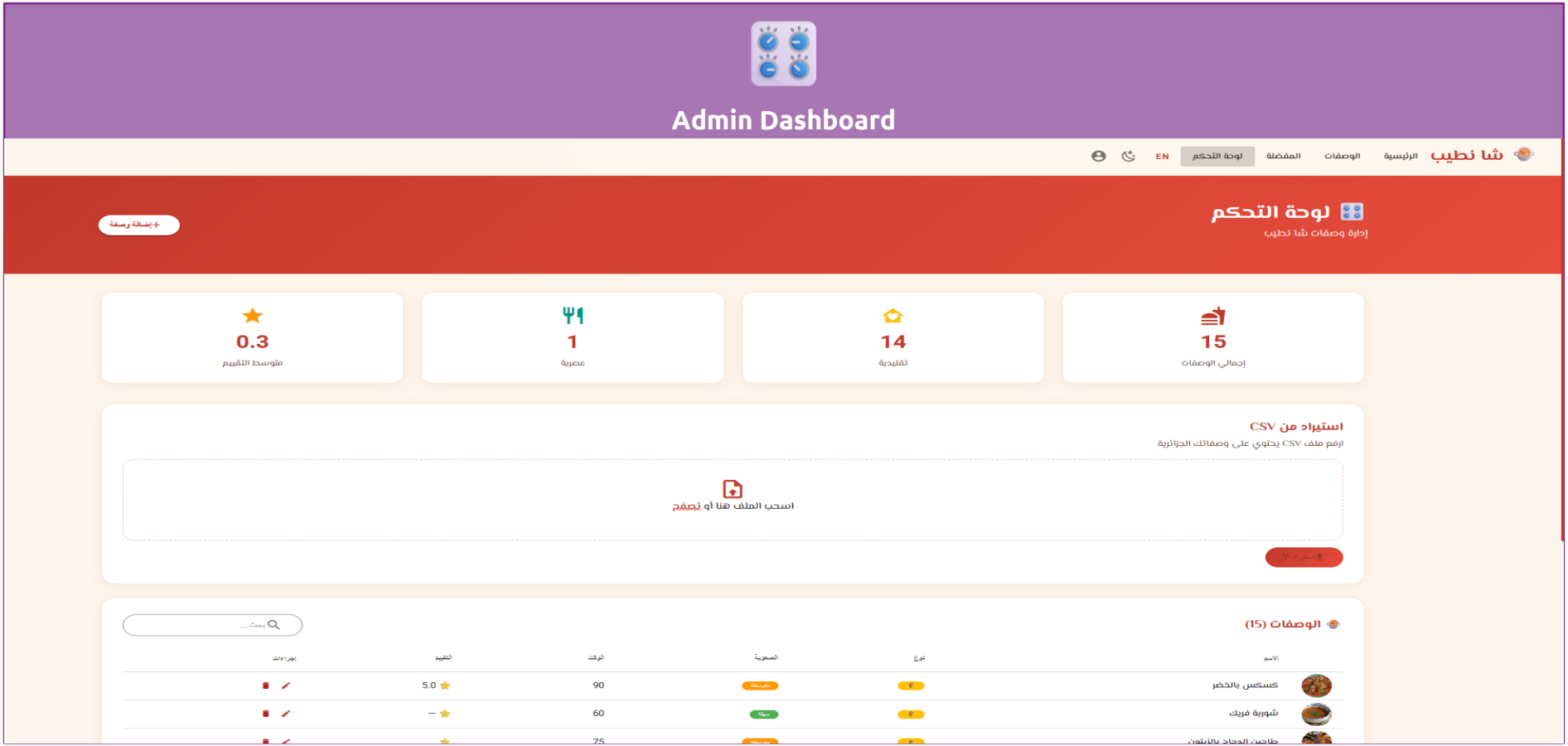


Part 4 : Results & Interpretation

Search Results

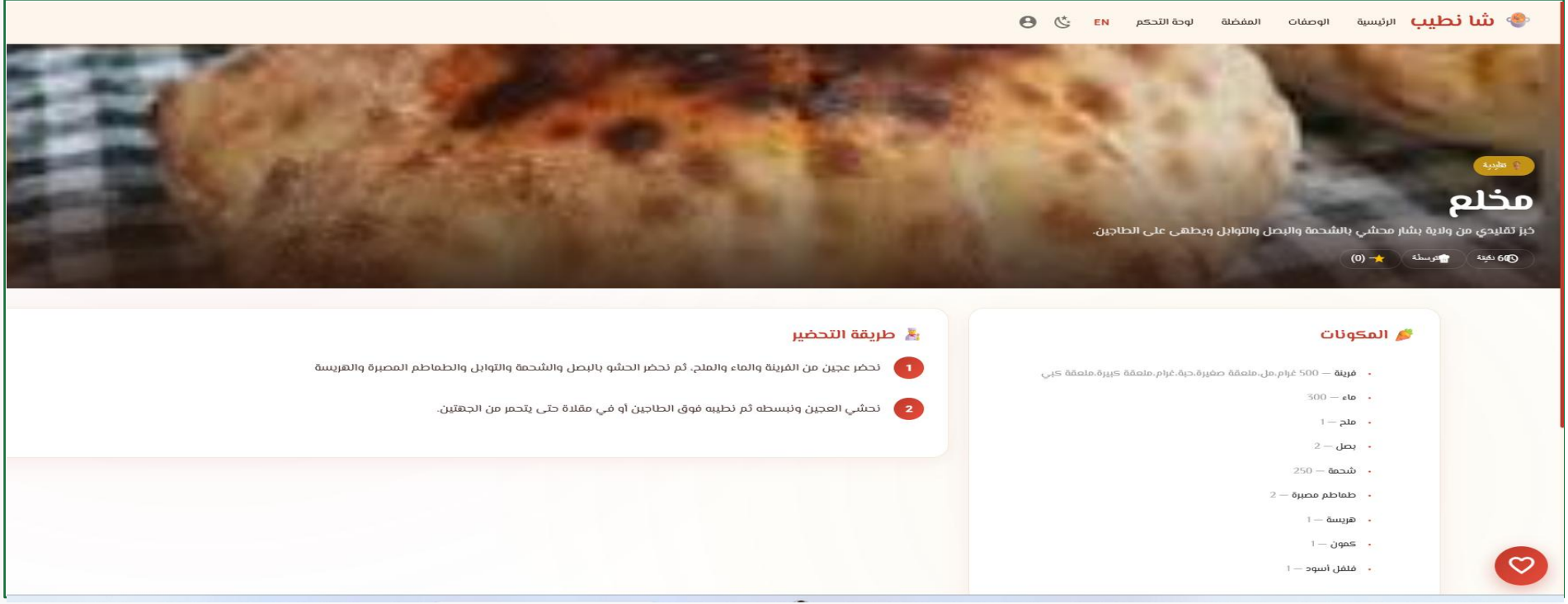


Part 4 :Results & Interpretation



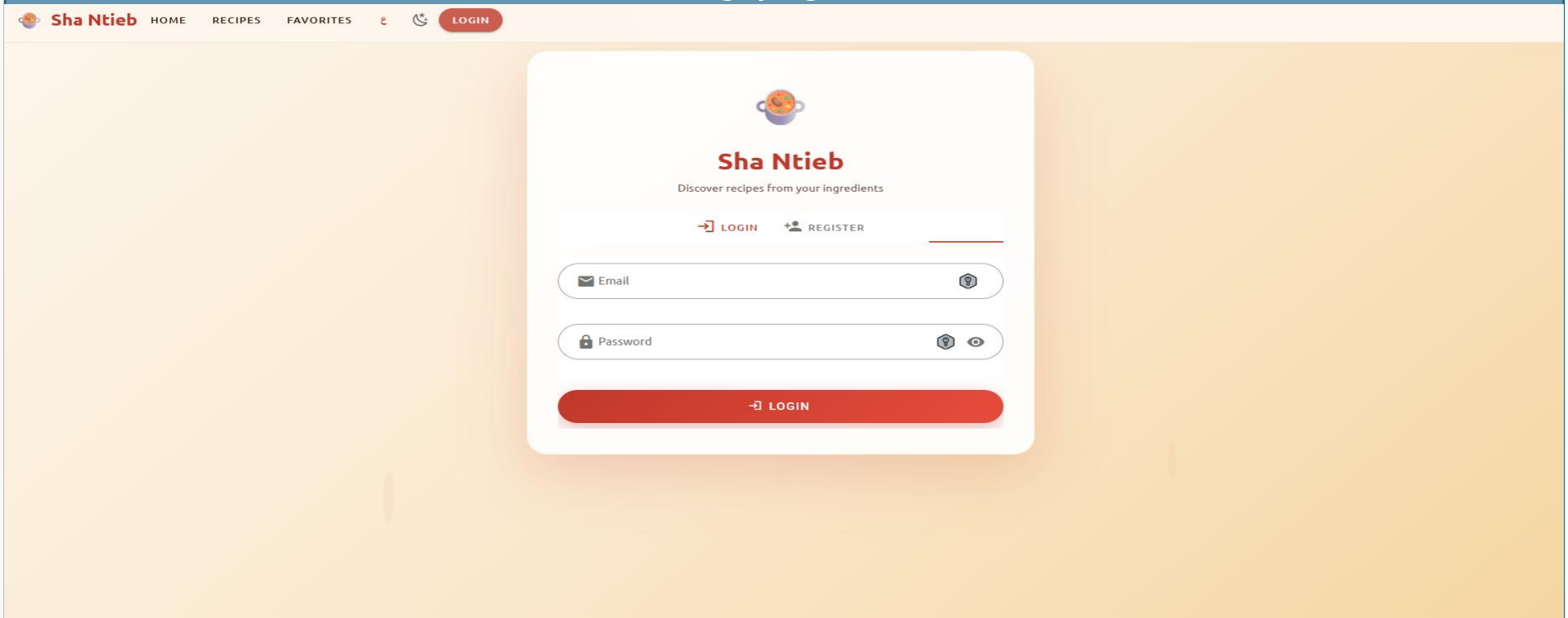
Part 4 : Results & Interpretation

Recipe Detail



Part 4 : Results & Interpretation

Login/Register



Part 4 :Results & Interpretation

Swagger API Docs

localhost:8000/docs

شا نطيب 1.0.0 OAS 3.1

/openapi.json

تطبيق اقتراح الوصفات الجزائرية

Auth

- POST /api/auth/register Register
- POST /api/auth/login Login

Recipes

- GET /api/recipes/ Get Recipes
- GET /api/recipes/{recipe_id} Get Recipe
- POST /api/recipes/search Search By Ingredients
- POST /api/recipes/rate/{recipe_id} Rate Recipe
- POST /api/recipes/ai-search Ai Search

Admin

- POST /api/admin/recipes Create Recipe
- PUT /api/admin/recipes/{recipe_id} Update Recipe
- DELETE /api/admin/recipes/{recipe_id} Delete Recipe
- POST /api/admin/import-csv Import Csv

Part 5 : Conclusion

📋 Summary of Work

- Built a full-stack web app for Algerian recipe recommendations
- Implemented 3-level AI engine: Simple Match + TF-IDF + Transformers
- Created Admin Dashboard with CSV import functionality
- Developed bilingual support (Arabic RTL + English LTR)
- Secured with JWT authentication and bcrypt password hashing

⚠️ Difficulties Encountered

- bcrypt version conflict with passlib library
- CSV encoding issues with Arabic text (BOM, UTF-8-sig)
- MySQL foreign key constraints during data deletion
- CORS configuration between FastAPI and Vue.js
- Arabic text normalization for AI matching accuracy

🚀 Areas for Improvement

- Add Flutter mobile application for iOS/Android
- Integrate GPT API for recipe explanation in Arabic
- Calculate nutritional values and calories per recipe
- Add voice search in Arabic using Speech Recognition
- Build community features for users to share recipes

شا نطيب

"واكتشف المطبخ الجزائري الأصيل ...أدخل مكوناتك"

Thank you for your attention